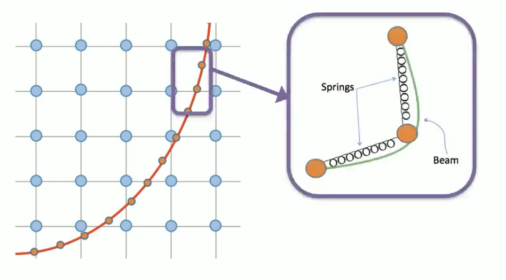
Immersed boundary method discretization

A power-consistent Eulerian–Lagrangian time step

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IBFE Tutorials



AI tools were used to assist with formatting, organization, and clarification of the instructor's original notes.

What this tutorial discretizes

Starting from a continuum IB model, we need discrete operators for:

1. incompressible flow on a fixed Cartesian grid;
2. elastic forces on a moving material mesh;
3. force spreading and velocity interpolation;
4. advancement of the coupled configuration.

The formulas below define one **first-order IMEX prototype**. They illustrate the coupling, not a unique IB time integrator.

IMEX: some terms are explicit and others implicit.

One coupling idea spans several codimensions

The immersed structure may be:

- a one-dimensional fiber in a two- or three-dimensional fluid;
- a two-dimensional surface in a three-dimensional fluid;
- a volumetric solid represented by finite elements.

The interaction formulas keep the same form; only the material measure and its quadrature change.

If $\mathcal{U}$ is the material domain,

$$\boldsymbol{f}(\boldsymbol{x}, t) = \int_{\mathcal{U}} \boldsymbol{F}(q, t)\delta\big(\boldsymbol{x} - \boldsymbol{\chi}(q, t)\big)\, dq.$$

Fibers, surfaces, and solids differ in the dimension and interpretation of dq.

Eulerian variables on a staggered grid

For clarity, use a uniform two-dimensional MAC grid with spacing h:

$$p_{i,j} \approx p(ih, jh), \qquad u_{i+\frac{1}{2},j} \approx u\big((i+\tfrac{1}{2})h, jh\big), \qquad v_{i,j+\frac{1}{2}} \approx v\big(ih, (j+\tfrac{1}{2})h\big).$$

- pressure is cell centered;
- each velocity component is face centered;
- the matching force component is stored on the same faces.

The discrete gradient G_h maps cells to faces, and the divergence D_h maps faces to cells. With compatible boundary treatment, $D_h = -G_h^*$.

Lagrangian geometry and quadrature

Let q_m label material points and let

$$\boldsymbol{X}_m^n = \boldsymbol{\chi}(q_m, t^n)$$

denote their current positions. Associate each sample or quadrature point with a material weight w_m.

Structure	Typical material measure	Weight
fiber	dq	$w_m \approx \Delta q$
surface	dA	surface quadrature weight
solid	dV	volume quadrature weight

w_m is a **quadrature measure**, not always a nodal spacing. In IBFE, quadrature points and structural nodes need not coincide.

Discrete interaction operators

Using Eulerian sites $\boldsymbol{x_i}$, cell/face weights $v_{\boldsymbol{i}}$, and a regularized kernel δ_h, define

$$(S_h[\boldsymbol{X}]\boldsymbol{F})_{\boldsymbol{i}} = \sum_m \boldsymbol{F}_m \, \delta_h(\boldsymbol{x_i} - \boldsymbol{X}_m) \, w_m, \qquad \text{force spreading,}$$

$$(J_h[\boldsymbol{X}]\boldsymbol{u})_m = \sum_{\boldsymbol{i}} \boldsymbol{u_i} \, \delta_h(\boldsymbol{x_i} - \boldsymbol{X}_m) \, v_{\boldsymbol{i}}, \qquad \text{velocity interpolation.}$$

On a uniform Cartesian grid, $v_{\boldsymbol{i}} = h^d$; for a MAC grid, each component uses the locations of its own faces.

S_h transfers a density per material measure to the fluid; J_h samples the fluid velocity at material points.

Matched coupling preserves discrete power

Equip the two grids with their weighted inner products:

$$\langle \boldsymbol{u}, \boldsymbol{f} \rangle_E = \sum_{\boldsymbol{i}} \boldsymbol{u_i} \cdot \boldsymbol{f_i} v_{\boldsymbol{i}}, \qquad \langle \boldsymbol{U}, \boldsymbol{F} \rangle_L = \sum_{m} \boldsymbol{U}_m \cdot \boldsymbol{F}_m w_m.$$

Substituting the definitions gives

$$\boxed{\langle \boldsymbol{u}, S_h \boldsymbol{F} \rangle_E = \langle J_h \boldsymbol{u}, \boldsymbol{F} \rangle_L}, \qquad J_h = S_h^*.$$

This is the discrete analogue of equal power exchanged between the fluid and structure.

A common kernel is not enough: **grids, weights, and inner products must be compatible**.

From the general framework to a concrete scheme

The coupling operators describe a *family* of immersed boundary methods. To make the algorithm concrete, it is useful to examine the classical fiber-based construction associated with Peskin's 1990s cardiac-flow work.

- The structure is a collection of material fibers.
- Elastic forces are assembled from targets, stretching, and bending.
- A regularized delta kernel couples the two grids.

The following choices explain the logic of one influential implementation. They are **not defining requirements** of every modern IB method.

Scope of the classical fiber discretization

The prototype considered here assumes:

- an incompressible Newtonian fluid;
- a uniform Cartesian Eulerian grid;
- periodic boundaries when an FFT-based fluid solver is used;
- immersed fibers described in Lagrangian coordinates;
- viscosity treated implicitly, with advection and fluid–structure coupling treated explicitly.

Later IB and IBFE schemes relax these choices through physical boundary conditions, adaptive grids, continuum solid models, and semi-implicit or fully implicit coupling.

Compute elastic forces from a discrete energy

If $E_h(\boldsymbol{X})$ is a discrete elastic energy, define force density by virtual work:

$$\delta E_h = -\sum_m \boldsymbol{F}_m \cdot \delta \boldsymbol{X}_m \, w_m, \qquad \boldsymbol{F}_m = -\frac{1}{w_m} \frac{\partial E_h}{\partial \boldsymbol{X}_m}.$$

Common contributions include:

- stretching or area/volume elasticity;
- bending and preferred curvature;
- active stresses or muscle models;
- target-point penalties.

For a prescribed target $\boldsymbol{Y}_m^n$,

$$\boldsymbol{F}_{\mathrm{targ},m}^n = k_{\mathrm{targ}}(\boldsymbol{Y}_m^n - \boldsymbol{X}_m^n).$$

Large penalty stiffness usually makes an explicit scheme more restrictive.

Example: conservative fiber tension

For equally spaced material labels, put tension on links:

$$\boldsymbol{d}_{m+\frac{1}{2}} = \frac{\boldsymbol{X}_{m+1} - \boldsymbol{X}_m}{\Delta q}, \qquad \boldsymbol{\tau}_{m+\frac{1}{2}} = \frac{\boldsymbol{d}_{m+\frac{1}{2}}}{|\boldsymbol{d}_{m+\frac{1}{2}}|},$$

$$\boldsymbol{t}_{m+\frac{1}{2}} = T\Big(|\boldsymbol{d}_{m+\frac{1}{2}}|\Big)\,\boldsymbol{\tau}_{m+\frac{1}{2}}, \qquad \boxed{\boldsymbol{F}_m^{\mathrm{str}} = \frac{\boldsymbol{t}_{m+\frac{1}{2}} - \boldsymbol{t}_{m-\frac{1}{2}}}{\Delta q}}.$$

The flux-difference form makes internal link forces equal and opposite. The constitutive law $T(\lambda)$ sets the reference stretch and stiffness.

Endpoint forces require the physical boundary conditions for the fiber.

Example: bending is a restoring force

For the energy

$$E_b = \frac{\kappa_b}{2} \int |\chi_{qq} - \chi^0_{qq}|^2 \, dq,$$

the interior force is

$$\boldsymbol{F}_b = -\kappa_b \, \partial_q^4 (\chi - \chi^0),$$

up to endpoint terms fixed by the boundary conditions. For a straight preferred configuration,

$$\boldsymbol{F}_{b,m} \approx -\kappa_b \frac{\boldsymbol{X}_{m-2} - 4\boldsymbol{X}_{m-1} + 6\boldsymbol{X}_m - 4\boldsymbol{X}_{m+1} + \boldsymbol{X}_{m+2}}{(\Delta q)^4}.$$

The minus sign is essential: the elastic force decreases the bending energy.

Classical fiber forces: three additive pieces

For material nodes $\boldsymbol{X}_m^n \approx \boldsymbol{\chi}(q_m, t^n)$ with spacing Δq, a common decomposition is

$$\boldsymbol{F}_m^n = \boldsymbol{F}_{\mathrm{targ,m}}^n + \boldsymbol{F}_{\mathrm{str,m}}^n + \boldsymbol{F}_{\mathrm{bend,m}}^n.$$

Target points give the penalty force

$$\boldsymbol{F}_{\mathrm{targ,m}}^n = k_{\mathrm{targ}}(\boldsymbol{Y}_m^n - \boldsymbol{X}_m^n).$$

With

$$(D_q \boldsymbol{X})_m = \frac{\boldsymbol{X}_{m+1} - \boldsymbol{X}_{m-1}}{2\Delta q}, \qquad \boldsymbol{\tau}_m = \frac{(D_q \boldsymbol{X})_m}{|(D_q \boldsymbol{X})_m|},$$

one illustrative Hookean stretching law is

$$\boldsymbol{F}_{\mathrm{str,m}} = k_{\mathrm{str}} D_q[(|D_q \boldsymbol{X}| - 1)\boldsymbol{\tau}]_m .$$

The link-tension form shown earlier is often preferable for making pairwise force balance transparent.

Classical fiber bending: sign and boundaries

For a straight preferred configuration, the interior bending force is

$$\boldsymbol{F}_{\mathrm{bend,m}} = -k_{\mathrm{bend}} \frac{\boldsymbol{X}_{m-2} - 4\boldsymbol{X}_{m-1} + 6\boldsymbol{X}_m - 4\boldsymbol{X}_{m+1} + \boldsymbol{X}_{m+2}}{(\Delta q)^4}.$$

The five-point stencil approximates $-k_{\mathrm{bend}}\partial_q^4 \boldsymbol{X}$ at interior nodes.

Closed fibers use periodic indexing. Open fibers require endpoint conditions consistent with the underlying bending energy.

Sign check: positive k_{bend} must produce a restoring force, so the discrete operator carries the leading minus sign.

A first-order IMEX fluid step

One pedagogical prototype treats advection and coupling explicitly and viscosity implicitly:

$$\rho\left(\frac{\boldsymbol{u}^{n+1}-\boldsymbol{u}^{n}}{\Delta t}+N_h(\boldsymbol{u}^{n})\right)=-G_h p^{n+1}+\mu L_h \boldsymbol{u}^{n+1}+S_h[\boldsymbol{X}^{n}]\boldsymbol{F}^{n}+\boldsymbol{g}^{n},$$
$$D_h \boldsymbol{u}^{n+1}=0.$$

Here L_h is the face-centered vector Laplacian and N_h is the chosen discrete convection operator.

Pressure appears as $-G_h p$ on the right-hand side (equivalently, $+G_h p$ on the left). The opposite sign would reverse the pressure force.

A component form of the classical fluid update

On a uniform grid, the same prototype may be written with finite-difference operators as

$$\rho\left[\frac{\boldsymbol{u}^{n+1}-\boldsymbol{u}^{n}}{\Delta t}+\mathcal{S}_h(\boldsymbol{u}^{n})\boldsymbol{u}^{n}\right]=-\boldsymbol{D}^{0}p^{n+1}+\mu\sum_{\alpha=1}^{d}D_{\alpha}^{+}D_{\alpha}^{-}\boldsymbol{u}^{n+1}+\boldsymbol{f}^{n},$$

$$\boldsymbol{D}^{0}\cdot\boldsymbol{u}^{n+1}=0.$$

The term $-\boldsymbol{D}^{0}p^{n+1}$ here and the term $-G_h p^{n+1}$ on the previous slide are the *same* pressure-gradient term written on two grids: G_h is the staggered (MAC) gradient, $\boldsymbol{D}^{0}$ its collocated centered-difference counterpart. Once $\boldsymbol{f}^{n}$ and the explicit convection term are known, the unknown velocity and pressure enter linearly.

On a periodic box, this structure makes FFT-based solvers natural. With physical boundaries or local refinement, other linear solvers are used.

Finite-difference operators in that update

For a scalar grid function ϕ and Cartesian unit vector $\boldsymbol{e}_\alpha$,

$$D^0_\alpha \phi(\boldsymbol{x}) = \frac{\phi(\boldsymbol{x} + h\boldsymbol{e}_\alpha) - \phi(\boldsymbol{x} - h\boldsymbol{e}_\alpha)}{2h},$$
$$D^+_\alpha \phi(\boldsymbol{x}) = \frac{\phi(\boldsymbol{x} + h\boldsymbol{e}_\alpha) - \phi(\boldsymbol{x})}{h}, \qquad D^-_\alpha \phi(\boldsymbol{x}) = \frac{\phi(\boldsymbol{x}) - \phi(\boldsymbol{x} - h\boldsymbol{e}_\alpha)}{h}.$$

One energy-conscious convection form is

$$\mathcal{S}_h(\boldsymbol{u})\phi = \tfrac{1}{2}\boldsymbol{u} \cdot \boldsymbol{D}^0\phi + \tfrac{1}{2}\boldsymbol{D}^0 \cdot (\boldsymbol{u}\phi).$$

On a staggered grid, the stencils differ; the design goal is unchanged: a compatible gradient/divergence pair with controlled kinetic-energy behavior.

One consistent explicit-coupling time step

Given $(\boldsymbol{u}^n, \boldsymbol{X}^n)$:

1. compute $\boldsymbol{F}^n = \mathcal{F}_h(\boldsymbol{X}^n, t^n)$;
2. spread $\boldsymbol{f}^n = S_h[\boldsymbol{X}^n]\boldsymbol{F}^n$;
3. solve the incompressible fluid step for $(\boldsymbol{u}^{n+1}, p^{n+1})$;
4. interpolate $\boldsymbol{U}^{n+1} = J_h[\boldsymbol{X}^n]\boldsymbol{u}^{n+1}$;
5. update $\boldsymbol{X}^{n+1} = \boldsymbol{X}^n + \Delta t\, \boldsymbol{U}^{n+1}$.

Every occurrence of the configuration in this prototype is evaluated at $\boldsymbol{X}^n$. Higher-order, midpoint, split, and implicit schemes use different—but deliberately consistent—time levels.

No slip is enforced through interpolation

At a material point q_m, the continuum kinematic equation is regularized as

$$\frac{d\boldsymbol{\chi}}{dt}(q_m, t) \approx \int_\Omega \boldsymbol{u}(\boldsymbol{x}, t)\, \delta_h\big(\boldsymbol{x} - \boldsymbol{\chi}(q_m, t)\big)\, d\boldsymbol{x}.$$

Its discrete counterpart is $\boldsymbol{U}_m = J_h[\boldsymbol{X}]\boldsymbol{u}$.

- the fluid velocity is averaged over the kernel support;
- the structure advances with that interpolated velocity;
- the same matched kernel spreads structural force back to the fluid.

At finite h, this is a regularized or weak no-slip relation rather than pointwise sampling of an exact interface trace.

Why bending and penalties can be stiff

For the discrete fourth derivative,

$$\lambda_{\max}(D_q^4) = \mathcal{O}((\Delta q)^{-4}).$$

Thus mesh-scale bending modes become rapidly stiffer as the Lagrangian mesh is refined.

What this does mean

- explicit elastic coupling may impose a severe Δt restriction;
- large target stiffness can have the same effect.

What it does not mean

- a resolved physical bending force itself grows like $(\Delta q)^{-4}$;
- IBFE or a penalty law automatically removes stiffness.

Time-step restrictions are scheme dependent

Explicit term	Representative scale	Important dependence
advection	$\Delta t \max_{i} \sum_{\alpha} \lvert u_\alpha \rvert / h_\alpha \lesssim C_{\mathrm{adv}}$	flux and time integrator
viscous diffusion	$\nu \Delta t / h^2 \lesssim C_\nu$	none with implicit diffusion
elastic coupling	no universal one-line bound	force law, grid, kernel, solver

There is no general rule that Δt is simply "inversely proportional to the applied force." Diagnose the unstable mode and the terms treated explicitly.

Resolve the interaction, not just the geometry

- **Too coarse:** gaps between interaction stencils can cause leakage and volume error.
- **Unnecessarily fine:** more work, redundant constraints, and possibly worse conditioning or high-frequency stiffness.
- **IBFE:** structural interpolation and interaction quadrature can be refined independently.

Rules such as $\Delta q \approx h/2$ are method- and kernel-dependent heuristics, not universal laws. Check leakage and convergence for the chosen discretization.

The kernel support, geometry, quadrature rule, and local Eulerian spacing all matter.

The kernel sets the interaction stencil

A common separable construction is

$$\delta_h(\boldsymbol{x}) = \frac{1}{h^d} \prod_{\alpha=1}^{d} \phi\Big(\frac{x_\alpha}{h}\Big).$$

Useful discrete identities include

$$\sum_{j\in\mathbb{Z}} \phi(r-j) = 1, \qquad \sum_{j\in\mathbb{Z}} (r-j)\phi(r-j) = 0.$$

The zeroth moment preserves total force and interpolates constants; the first moment centers spreading and supports torque/linear-field reproduction.

Tensor-product kernels are efficient and grid aligned; they are not exactly invariant under rotation.

Example: a four-point-support cosine kernel

One simple compactly supported choice is

$$\phi(r) = \begin{cases} \dfrac{1}{4}\left(1 + \cos\dfrac{\pi r}{2}\right), & |r| \le 2, \\ 0, & |r| > 2, \end{cases} \qquad \delta_h(\boldsymbol{x}) = \frac{1}{h^d} \prod_{\alpha=1}^{d} \phi(x_\alpha/h).$$

It is symmetric and compactly supported: in each coordinate, only grid sites within $2h$ contribute.

"Four-point support" describes the one-dimensional stencil width. In d dimensions, the tensor-product support contains up to 4^d grid sites per Lagrangian point.

Note: $\phi_{\cos}$ satisfies the zeroth moment and $\sum_j \phi_{\cos}(r-j)^2 = 3/8$ but *not* the first-moment condition ($\max |M_1| \approx 0.021$); see deck 1-3.

Verification checks for an implementation

Operator checks

- $D_h = -G_h^*$ under the chosen boundary treatment;
- $J_h = S_h^*$ in weighted inner products;
- constant and linear-field kernel tests;
- total force and torque after spreading.

Solution checks

- incompressibility and solver residuals;
- energy or power balance when applicable;
- independent refinement in h, quadrature/mesh spacing, and Δt;
- sensitivity to domain size and boundary conditions.

Refining every discretization parameter together can hide the source of an error.

What convergence requires

For a conventional regularized-delta IB discretization:

- the interaction layer has width $\mathcal{O}(h)$;
- singular forces and pressure/traction jumps are therefore smoothed at finite h;
- interfacial quantities are often less accurate than smooth fields away from the interface.

$\delta_h \to \delta$ distributionally is **not by itself** a convergence proof. The fluid scheme, time integrator, material mesh or quadrature, and coupling must be stable and refined consistently.

From the prototype to modern IB solvers

The same interaction-operator viewpoint extends to:

- block-structured adaptive mesh refinement;
- semi-implicit or fully implicit fluid–structure coupling;
- finite-element descriptions of volumetric solids (IBFE);
- constrained-body and sharp-interface variants;
- scalable multilevel linear and nonlinear solvers.

Uniform grids, periodic boxes, FFT solvers, and explicit coupling are choices of particular implementations—not defining assumptions of the immersed boundary formulation.

Classical choices and modern extensions

Classical fiber prototype

- fiber structures;
- spring and beam elasticity;
- commonly explicit coupling;
- uniform, often periodic grid;
- FFT solver on a periodic box.

Modern IB/IBFE examples

- fibers, surfaces, and volumetric solids;
- FE continuum and active-material laws;
- explicit through monolithic coupling;
- physical boundaries and adaptive grids;
- scalable multigrid and Krylov solvers.

The enduring idea is the Eulerian–Lagrangian interaction pair. The structure model, grid hierarchy, and time integrator can all change around it.

Roadmap from the classical method

What the prototype teaches

- moving geometry without remeshing;
- forces and kinematics transferred by a common kernel;
- a transparent sequence for one coupled time step.

What motivates later methods

- stiffness under explicit elastic coupling;
- reduced interfacial accuracy from regularization;
- the need for continuum solids, adaptivity, and robust solvers.

These limitations lead naturally to implicit IB schemes, immersed interface variants, adaptive solvers, and the IBFE formulation developed later in this tutorial series.

Discretization checklist

1. Put Eulerian variables and discrete differential operators on compatible locations.
2. Treat material weights as quadrature measures and compute forces with the correct sign.
3. Construct S_h and J_h as weighted adjoints so coupling preserves power.
4. State the time levels and implicit/explicit split unambiguously.
5. Test conservation, leakage, stability, and convergence independently.

References and further study

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